

No. of Functional Features Included in the Solution

Date	15 July 2026
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

This document lists all the functional features included in Rising Waters. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Flood Risk Prediction Engine	XGBoost classifier predicts flood / no-flood from 5 rainfall and cloud-cover features	core/predictor.py, models/	Done	Core
2	Rainfall Input Form	Collects Cloud Cover, Annual Rainfall, and three seasonal rainfall values	templates/index.html	Done	Core
3	Client-Side Validation	Checks required fields, numeric values, and realistic ranges before submission	static/js/script.js	Done	Core
4	Server-Side Validation	Re-validates all 5 fields server-side to prevent bad data or crashes	core/validation.py, app.py	Done	Core
5	Result Visualization	Circular probability gauge plus a clear flood-risk / safe verdict page	templates/chance.html, nochance.html	Done	Core
6	Animated Landing Page	Project introduction with an animated SVG rain-gauge visual	templates/home.html	Done	Nice-to-have
7	Model Training Pipeline	Auto-detects real rainfall data or falls back to synthetic data, then trains and saves the model	train.py	Done	Core
8	Model Artifact Storage	Persists the trained classifier and scaler as joblib files for fast runtime loading	models/floods.save, transform.save	Done	Core

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	7
Additional / Nice-to-Have Features	1
Features Tested & Verified	8

Feature Category Breakdown

S. No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Landing page, input form, client-side validation, result pages
2	Backend / Logic	3	Prediction engine, server-side validation, training pipeline
3	Database / Storage	1	Model artifact storage (joblib flat files)
4	API / Integration	0	None yet — NASA POWER rainfall API integration planned for Phase 2
5	Security / Authentication	0	Not applicable — public, anonymous access protected only by input validation